# WONG, Hok Fong

Intern, Summer@EPFL; BSc in Computer Science, CUHK

**Tel.**: (+852) 63165455
**Email**: wonghf222 [at] gmail.com
**Homepage**: swonghf.github.io/
**GitHub**: github.com/SWongHF


## Academic Interests

Machine Learning and Theory; Optimization; Design and Analysis of Algorithms; Theoretical Computer Science; Atmospheric Science

## Education

**The Chinese University of Hong Kong, Hong Kong** [**Cumulative GPA: 3.803 / 4.0** | **Major GPA: 3.916 / 4.0**] *Aug 2022 – Jun 2026*
**Major**: ***B.Sc. in Computer Science*** with **First-Class Honors** (Focus: Algorithms and Complexity) Academic Transcript
**Minors**: ***Mathematics; Earth and Environmental Sciences*** (Focus: Atmospheric Science) **(151 Units = ~302 ECTS** [Ref]**)**
**Engineering Enhancement Program:** ***Engineering Leadership, Innovation, Technology and Entrepreneurship Stream*** (**ELITE Stream**)

## Publication

**H. F. Wong**, H.-T. Wai, C.-Y. Yau, "Large Stepsizes Federated Learning on Logistic Regression with Linearly Separable Data: The Case of Heterogeneous Devices," *The 65th IEEE Conference on Decision and Control* (**accepted**), 2026. **(Theory Paper)**

## Work Experiences

**(Ongoing) Junior Research Assistant – Dept. of Earth and Environmental Sciences, CUHK** *Sep 2026 – Aug 2027*
- ***Atmospheric Chemistry Modeling and Climate Group***. Actual responsibilities TBD. Tentatively on AI/ML for atmospheric chemistry, through ***developing a global anthropogenic fugitive dust emission inventory with multi-source data*** with the aim of publishing a paper as main author.
- ***Official teaching assistant of the undergraduate/research postgraduate course EESC4550/5550*** – Practical Atmospheric Modeling (2026 Fall), involved in course content development (Linux HPC Computing, AI/ML Toolbox for Atmospheric Chemistry), design of assignments and lab exercises (Modeling Stratospheric Ozone Dynamics, GEOS-Chem Tutorials). (Available upon request.)
- (Upcoming) Teaching assistant of the taught postgraduate course AISC5010 – Basic Programming for AI (2027 Spring).
- **Supervisor**: **Prof. Shixian ZHAI**, Department of Earth and Environmental Sciences, CUHK.

**Summer Research Fellow – Summer@EPFL 2026, Switzerland** *Jun 2026 – Aug 2026*
- Selected from 7,000+ global applicants (***1.2% acceptance rate***), with the focus on ***optimization for machine learning (theoretical analysis)***.
- On ***high-probability guarantees of SGD with linear warm-up*** under $(H_0, H_1)-$ $\mathrm{smoothness\ and\ aiming\ condition, targeting\ ICML\ 2027;\ the\ result\ directly\ addresses\ two\ major\ weaknesses\ of\ the\ state-of-the-}$ $\mathrm{art\ analysis:\ impractical\ dependence\ on\ } f^*$ and the interpolation requirement. [Slides]
- **Supervisor**: **Prof. Martin JAGGI** and **Dr. El Mahdi CHAYTI**, Machine Learning and Optimization Laboratory (MLO), EPFL.

**Research Student – CUHK Undergraduate Summer Research Internship (2024, 2025)** *Jun 2024 – May 2026*
- Conducted research in modern optimization methods tailored for ***large-scale machine learning***, focusing on stochastic methods for nonconvex problems.
- Analyzed theoretical properties such as ***convergence guarantees, and complexity bounds*** under smoothness and/or bounded gradient assumptions.
- Awarded ***research grants*** from Faculty of Engineering, with a total amount of ***HKD $28,000***.
- **Supervisor: Prof. Hoi-To WAI**, Department of System Engineering and Engineering Management, CUHK.

**Curriculum Designer – CUHK & Jockey Club "AI for the Future" Project** *Mar 2023 – Dec 2024*
- Spearheaded the ***creation of engaging courseware*** introducing the knowledge of Artificial Intelligence (AI) to thousands of secondary school students.
- Collaborated with professors and post-doctoral researchers to design curriculum content aligned with the ***first AI syllabus launched for junior secondary education in Hong Kong***. [Website]
- Received stipends with a total amount of ***HKD $12,000*** for the technical quality and the education impact of the work conducted.
- **Supervisor**: **Prof. Mei Ling Helen MENG**, **Dr. Xing XING**, Stanley Ho Big Data Decision Analytics Research Centre, CUHK.

## Selected Projects

**A more detailed list of projects, essays, and codes are available:** [Link]

**Theoretical Analysis of Large-Stepsize Gradient Descent on Multinomial Logistic Regression** *Fall 2025, Spring 2026*
- ***Graduation thesis challenging conventional smoothness-based convergence limits for large-stepsize Gradient Descent***. [PDF]
- Rigorous theoretical analysis of the ***optimization dynamics of multinomial logistic regression with separable data***, delivering its ***first non-asymptotic convergence bound***. Precisely quantified that ***loss and phase transition time scale only logarithmically with the number of classes***.
- Extended analysis to ***FedAvg with large stepsizes and heterogeneous local updates*** where heterogeneity vanished asymptotically. Also characterized convergence on ***non-separable data***: adaptive stepsizes guarantee monotonic $O(1/t)$ convergence, constant-stepsize cycles exist for $K \geq 3$ classes.

**On Quantization-Aware Training Algorithms and Stronger Convergence Guarantees of** `ASkewSGD` *Summer 2024, 2025*
- ***Research project*** focused on ***improving theoretical guarantees*** of the `ASkewSGD` algorithm for ***quantizing deep neural networks (DNNs)***.
- Eliminated the need for convexity, Lipschitz continuity and differentiability requirements commonly adopted in prior work via Stochastic Approximation.
- Demonstrated `ASkewSGD`'s superiority over benchmarks (`BinaryConnect`, `ProxQuant`) in tasks such as synthetic data (`TwoMoons`) and image classification (`MNIST`, `CIFAR-10`) with low-bit quantization for resource-efficient DNN deployment. We publicized reproducible experiments on GitHub.

**Frequency Estimation in Strict-Turnstile Data Streams with $L_1$ $\alpha$-Bounded Deletions** *Spring 2026*
- Extended four ***classical streaming algorithms*** (`Misra-Gries`, `SpaceSaving`, `Count-Min Sketch`, `Count-Sketch`) to the strict turnstile model with the $L_1$ $\alpha$-bounded deletion property, proving **tighter error bounds that scale with** $\alpha$ ***instead of stream length***.
- Conducted ***extensive experiments on Zipfian, uniform, and real-world CAIDA network traces (sliding window and idle timeout streams)***.
- Proposed a novel algorithm, ***Dyadic Count-Min Hierarchy with Adaptive Counter Cache***, a hybrid that ***combines logarithmic space sampling with a small deterministic cache***, preserving counter-based accuracy for the heaviest items while remaining space efficient for large $\alpha$.
- Project available on GitHub.

**First-Order Methods in Online Learning and Online Convex Optimization** *Fall 2024*
- ***Studied and implemented key algorithms*** such as Online Gradient Descent (`OGD`), Follow-the-Regularized-Leader (`FTRL`), and Online Mirror Descent (`OMD`), analyzing their ***regret bounds and convergence behavior under adversarial settings*** with strong convexity and smoothness assumptions.
- Conducted empirical evaluations comparing algorithmic performance across synthetic and real-world datasets, focusing on regret minimization and computational efficiency for ***online learning / online convex optimization***. The project is available on GitHub.

**A Practical Test Set Bound in Binary Classification Theory** *Fall 2023*
- Reconstructed and analyzed key proofs by John Langford to deepen understanding of ***generalization bounds for binary classification based on merely the empirical risk and size of the training set***, grounded in principles from statistical learning theory.
- Implemented simulations to apply the proposed bounds under given conditions of proposed model performance. The project is available on GitHub.

## Selected Academic Awards

**Best Project Award - UG Summer Research Internship 2025 (Faculty of Engineering, CUHK)** *2025*
- ***Ranked top 10% among all undergraduate summer research projects***, based on originality, analytical rigor, and professional delivery of the project "Revisiting `ASkewSGD`: New Theoretical Guarantees for Quantization-Aware Deep Neural Network Optimization". [Web Archive]

**Andrew C. Yao Fellowship – HKD $50,000** *2024*
- Awarded to ***only 2 outstanding CSE undergraduates*** annually based on ***academic distinction and exceptional extracurricular achievements***.

- ***Highest distinction in CSE department, named in honor of Professor Andrew Chi-Chih Yao (Turing Award, 2000).*** [Hall of Fame]

**ELITE Stream Scholarship – HKD $20,000** *2024, 2026*
- Awarded to students of the Engineering Leadership, Innovation, Technology and Entrepreneurship Stream with ***outstanding performance in recognized engineering-discipline courses with challenging coursework***. [Official Description]

**CUHK Vice-Chancellor's Scholarships for Excellence – HKD $50,000** *2023*
- One of the only ***8 first-year undergraduate students selected university-wide*** for ***exceptional academic and distinguished extracurricular achievements***, recognizing top-tier students with outstanding potential. [Official Press Release]

**HKSAR Government Talent Development Scholarship – HKD $10,000** *2023*
- For the exceptional achievements in the ***innovation, science and technology area (non-academic)***. [Official Description]

**Dr. Stanley Ho Medical Development Foundation Scholarship – HKD $18,000/yr. (renewable)** *2022, 2023, 2024, 2025*
- Awarded to CUHK students from Macao with ***excellent academic performance*** (offered by Morningside College, CUHK). [College News]

**Dean's List (Faculty of Engineering, CUHK)** *2023, 2024, 2025, 2026*
- Awarded to Engineering students with ***excellent academic performance (Top 10%)***.

## Competitions

**Huawei ICT Competition (Hong Kong Regional, Final Round) - Cloud Track** *Nov 2022 – Dec 2022*
- ***Potential Award*** with ***HKD $1,500***, for knowledge and skills related to cloud computing and Huawei's cloud technologies.

**International Collegiate Programming Contest (ICPC), Asia Macau Region** *May 2021, Apr 2022*
- Awarded ***Outstanding High School Team*** with ***MOP $1,000*** (for both years).

**Macao Olympiad in Informatics** *2018, 2019, 2020, 2021, 2022*
- Consistent remarkable performance with an accumulation of ***4 Gold Medals* (Top 5%) and *1 Silver Medal*** from the first participation onward. [Link]
- Designed, implemented, and tested ***efficient algorithms and data structures*** in C++ to solve problems under time and memory constraints with ***strong logical reasoning and programming proficiency***.
- Competed in ***National Olympiad in Informatics* (CCF NOI '19)**, ***International Olympiad in Informatics* (IOI '21, '22)**, ***Team of Macao***. [Link]

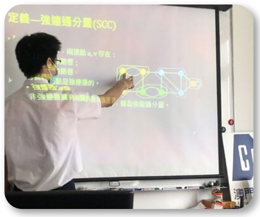
## Leadership & Extracurriculars

**Vice Chairman and Secretary General - The Macao Association for Competitive Programming** *Apr 2026 – Apr 2029 (Est.)*
- Government recognized registered association, with the goal of cultivating students' algorithmic thinking through seminars, invited talks, and contests.
- Co-organized the "HKACE 45th Anniversary Cup – HKOI 2026 Asia Invitational" in the Macao Region with Macao Computer Society in May 2026.
- Official Website: [Link]; Instagram page: [Link].

**Participant - Greece Service-Learning Trip (Organized by Morningside College, CUHK)** *June 2024*
- Engaged with non-governmental organizations (Doctors of the World, etc.) with the focus on the ***Greek NHS and healthcare access for refugees***.
- Performed ***critical evaluations*** related to ***operation structures of NGOs*** and how ***challenges related to refugees and asylum seekers*** are tackled.
- Authored a ***comprehensive 10,000-word field journal*** and submitted a formal ***evaluation report*** proposing actionable improvements to healthcare delivery and NGO coordination. Related documents are available here and here.
- **Reference**: **Morningside College, CUHK** (College News Highlight).

**Secretary - The 12th Morningside College Student Union (Executive Council)** *Mar 2023 – Feb 2024*
- Maintained accurate meeting minutes and digitalized organizational records, ensuring transparency and continuity across student-led operations.
- Excelled in ***drafting activity proposals and managed complex logistics***.
- Led initiatives that fostered ***inclusivity and cross-cultural engagement*** that enhanced the well-being of Morningside College's diverse and international student community.
- ***Organized events and activities*** including Graduation Photo Day, Mid-Autumn Festival, Halloween Party, Cultural Fair, Poon-Choi Dinner ***under strict budget constraints***.
- **Reference**: **Morningside College, CUHK** (Event 1 Event 2 Event 3&4 Event 5).


**Member - Macao Computer Society** *Sep 2020 – Present*
- ***Provided comprehensive training programs*** online and offline for the Macao team at the International Olympiad in Informatics (IOI) at year 2022, 2023. [Selected Session's Slides]
- ***Designed the banner and served as an invigilator for Macao Olympiad in Informatics (MOI) in 2023,*** as a ***member of the organizing committee***. [Banner] [Ceremony]
- Served as ***invigilator for Macao Olympiad in Artificial Intelligence (MOAI) in 2026***.
- ***Proposed 2 original and challenging ICPC-styled competitive programming problems to the Joint-School Invitational Contest*** in 2022, designed to test algorithmic thinking and problem-solving skills among secondary school participants, fostering a sense of community and healthy competition.
  - **Problem 1** **(Prime Maze)**: Observations in number theory and search algorithms, requiring solvers to identify mathematical patterns to avoid inefficient brute-force solutions. [Editorial Available]
  - **Problem 2** **(Matchmaker)**: Basic number theory with (advanced) data structure implementations.
- **Reference**: **Mr. Kuo-Meng SUN** (Vice-executive director of MCS).



## Technical Skills

**Language:** Cantonese [Native], Mandarin [Native], English [Proficient] (IELTS Academic 7.5, L-9.0 R-8.5 W-6.5 S-6.5, Test date: Jun 2024).
**Programming:** C, C++, Java, Python{PyTorch, pandas, openpyxl, scikit-learn, Matplotlib, Django}, OCaml, Prolog, JavaScript, HTML, CSS.
**Tools:** SQL, LaTeX, Git, WandB, Linux, GEOS-Chem.
**IT Certifications:** LPI Linux Essentials 010-160 (Score: 800/800); CCF Assessment of Programming Proficiency Level for Secondary Students – Level 7.
**Competitive Programming:** Efficient and Optimized Computations, Algorithmic Thinking & Problem-Solving, Pedagogical Skills.

## Courseworks

**Mathematics:** Single/Multi-variable Calculus, Linear Algebra, Discrete Mathematics, Probability and Statistics, Game Theory, Ordinary Differential Equations, Optimization Theory, Mathematical Foundations for Data Analytics.
**Computer Theory:** Data Structures and Algorithms, Formal Languages and Automata Theory, Principles of Programming Languages, Computational Geometry (Postgraduate-Level), Algorithms for Big Data (Postgraduate-Level).
**AI&ML:** Fundamentals of Artificial Intelligence, Fundamentals of Machine Learning, Numerical Optimization.
**Systems:** Digital Logic and Systems, Computer Organization, Software Engineering, Operating Systems.
**Earth Science:** Climate System Dynamics, Solid Earth Dynamics, Ecological Climatology, Cloud Dynamics, Atmospheric Chemistry and Modeling.

**Last updated: Aug 31, 2026**